

PROJECT PROPOSAL



PARKING MANAGEMENT SYSTEM

TRIDSADATE GREATSANGTHONG
NANTAWUT PANAN
THANATHIP PITAKSIN
PATTAKORN JAIJIT

BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING

MAE FAH LUANG UNIVERSITY
2025

© COPYRIGHT BY MAE FAH LUANG UNIVERSITY

PARKING MANAGEMENT SYSTEM

**TRIDSADATE GREATSANGTHONG
NANTAWUT PANAN
THANATHIP PITAKSIN
PATTAKORN JAIJIT**

**A COMPUTER ENGINEERING PROJECT SUBMITTED TO
MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING**

**MAE FAH LUANG UNIVERSITY
2025**

© COPYRIGHT BY MAE FAH LUANG UNIVERSITY

PARKING MANAGEMENT SYSTEM

TRIDSADATE GREATSANGTHONG

NANTAWUT PANAN

THANATHIP PITAKSIN

PATTAKORN JAIJIT

THIS SENIOR PROJECT HAS BEEN APPROVED TO BE A PARTIAL
FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING 2025

EXAMINING COMMITTEE



..... ADVISOR

(Asst.Prof. Mahamah Sebakor)



..... COMMITTEE

(Asst.Prof. Surapol Vorapatratorn)



..... COMMITTEE

(Mr. Noppakan Pinit)

ACKNOWLEDGEMENT

This project will not be successful without the kind support of the advisor, committees and all supportive persons. We would like to thank everyone for their advice and help. We are also grateful to our parents and friends for their continuous encouragement throughout this project. This accomplishment would not have been possible without them. Thank you.

Tridsadate Greatsangthong
Nantawut Panan
Thanathip Pitaksin
Pattakorn Jaijit

Title	Parking Management System
--------------	---------------------------

Author Tridsadate Greatsangthong
Nantawut Panan
Thanathip Pitaksin
Pattakorn Jaijit

Degree	Bachelor of Engineering (Computer Engineering)
---------------	---

Supervisory Committee	Asst.Prof. Mahamah Sebakor	Advisor
	Asst.Prof. Surapol Vorapatratorn	Committee
	Mr. Noppakan Pinit	Committee

Abstract

This senior project presents the development of a “Parking Management System” designed to address the growing challenges of vehicle congestion and inefficient parking space utilization in modern environments.

The proposed system leverages existing CCTV infrastructure combined with AI-based detection technologies Ultralytics YOLO26 to provide real-time monitoring of parking slot availability. The system also has License Plate Recognition (LPR) and Facial Recognition to enhance security and automate data collection vehicle entry and exit parking.

The expected outcomes to improved parking space utilization and enhanced security for vehicles and users.

Keyword: Parking Management System, YOLO26, License Plate Recognition, Facial Recognition, CCTV

CONTENTS

ACKNOWLEDGEMENT	iii
Abstract	v
LIST OF TABLES	vii
LIST OF FIGURES	viii
CHAPTER 1 INTRODUCTION	1
1.1 Background and Rational	1
1.2 Objectives	1
1.3 Scope	2
1.4 Methodology	2
1.5 Plan	3
1.6 Expected Result	3
1.7 Place and Equipment	4
CHAPTER 2 LITERATURE REVIEW	5
2.1 Related Research	5
2.2 Related Technology	5
2.3 Related Work	11
CHAPTER 3 ANALYSIS AND DESIGN	14
3.1 System Diagram	14
3.2 Use Case Diagram	17
3.3 UI Design	19
CHAPTER 4 CONCLUSION	22
4.1 Conclusion	22
REFERENCES	23

LIST OF TABLES

Table	Page
Table 1 Working Plan	3

LIST OF FIGURES

Figure	Page
Figure 1	5
Figure 2	6
Figure 3	7
Figure 4	7
Figure 5	8
Figure 6	8
Figure 7.1	9
Figure 7.2	9
Figure 8	10
Figure 9	10
Figure 10	11
Figure 11	16
Figure 12	17
Figure 13	17
Figure 14	18

Figure	Page
Figure 15	19
Figure 16	19
Figure 17	20
Figure 18	20
Figure 19	20
Figure 20	21
Figure 21	21

CHAPTER 1

INTRODUCTION

1.1 Background and Rational

Nowadays, more people prefer driving their own cars for comfort and privacy, leading to a steady increase in car ownership. This rise in vehicles causes more traffic and makes it much harder to find parking, especially in crowded or indoor parking lots. Traditional or manual parking management forces both drivers and staff managers to locate available spaces manually. This unorganized approach results in a significant waste of time, increased energy consumption, and unnecessary stress.

To solve this problem, this project proposes development of a "Parking Management System." Included with Camera detection system and existing CCTV monitoring infrastructures. The system delivers real time parking availability without the need for expensive ground hardware detection. As a developer, the core objective is to create a seamless, high-performance ecosystem accessible to all users.

At lease, this project aims to improve and optimize enterprise solutions offering a scalable, parking management system platform that saves time, reduces energy waste, and optimizes space utilization effectively.

1.2 Objectives

1.2.1 To maximize available parking, reduce unused spaces and much efficient turnover.

1.2.2 To implement a centralized data management system for recording and analyzing parking usage patterns in future planning.

1.2.3 Enhance security parking for unexpected scenarios to users.

1.3 Scope

1.3.1 General User

- 1) Check parking locations to see where buildings are empty or not.

1.3.2 Staff

- 1) Monitoring for checking available cars for parking.
- 2) In case have an emergency with parking staff can manage to resolve the problem with any scenario.

1.3.3 Admin

- 1) Create and maintain the system, manage staff, and collect the data to store in the databases.

1.4 Methodology

1.4.1 Proposal preparation and defense

1.4.2 Literature review and requirement gathering

1.4.3 System analysis

1.4.4 System design

1.4.5 Senior project document preparation

1.4.6 System development

1.4.7 System testing

1.4.8 Presenting the system

1.5 Plan

Table 1. Working Plan

	Senior Project 1					Senior Project 2				
Plan/Month	Jan	Feb	Mar	Apr	May	Aug	Sep	Oct	Nov	Dec
Proposal preparation and defense										
Literature review and requirement gathering										
System analysis										
System design										
Senior project document preparation										
System development										
System testing										
Presenting the system										

1.6 Expected Result

1.6.1 Parking spaces are utilized effectively with better organization.

1.6.2 Reduced parking traffic congestion.

1.6.3 Security and support for parking will be upgraded.

1.7 Place and Equipment

1.7.1 Place

Mae Fah Luang University (Test at E4 Building)

1.7.2 Hardware

- 1) Computer/Laptop
- 2) Mobile
- 3) CCTV

1.7.3 Software

- 1) Visual Studio Code
- 2) PostgreSQL and pgAdmin (Database)
- 3) [Node.js](#) (Back-end)
- 4) [vue.js](#) (Front-end web)
- 5) Python, Ultralytics YOLO26 (Camera detection)
- 6) Figma (UI Design)

CHAPTER 2

LITERATURE REVIEW

2.1 Related Research

2.1.1 How does Ultralytics YOLO26 enhance parking management systems?

Ultralytics YOLO26 greatly enhances parking management systems by providing real-time vehicle detection and monitoring. This results in optimized usage of parking spaces, reduced congestion, and improved safety through continuous surveillance. The Parking Management System enables efficient traffic flow, minimizing idle times and emissions in parking lots, thereby contributing to environmental sustainability. For further details, refer to the parking management code workflow.

2.1.2 What is LPR?

License Plate Recognition (LPR) cameras, commonly known as license plate reading cameras, are more than just standard surveillance cameras. They are intelligent systems that integrate AI and advanced image processing technologies to enhance security and improve the efficiency of various management systems.



Figure 1 License Plate Recognition

2.1.3 Facial Recognition

Facial recognition is a specialized biometric technology that utilizes artificial intelligence (AI) to identify or verify an individual's identity by analyzing patterns based on their facial features. Unlike standard computer vision (CV) tasks that might simply classify an image, facial recognition systems employ complex mathematical mapping to interpret the unique geometry of a human face. This technology has rapidly evolved from theoretical research to becoming a ubiquitous tool in machine learning (ML), powering everything from smartphone security to advanced surveillance and streamlined customer experiences.

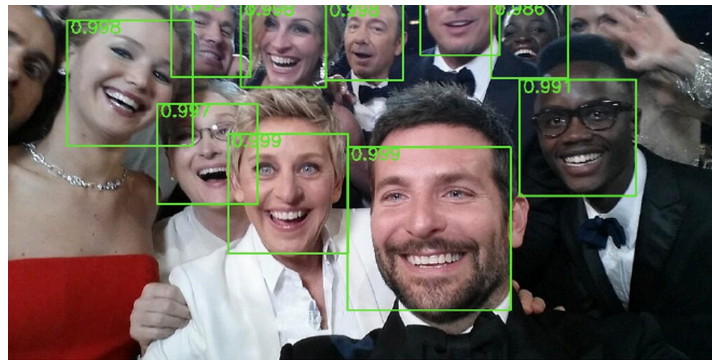


Figure 2 Facial recognition

2.2 Related Technology

2.2.1 Visual Studio Code

Visual Studio Code is one of the popular text editors used by professionals and recommended to new coders. Being one of Microsoft's renowned software, VS code is free to use, open-source, and compatible with Windows, Linux, and macOS



Figure 3 Visual Studio Code

2.2.2 Figma

Figma is a web-based design tool used for creating user interfaces (UI) and user experiences (UX) for apps and websites. It works directly in your browser and allows real-time collaboration.



Figure 4 Figma

2.2.3 [Vue.js](#)

Vue.js is a JavaScript framework for building user interfaces. It builds on top of standard HTML, CSS, and JavaScript and provides a declarative, component-based programming model that helps you efficiently develop user interfaces of any complexity. Vue is designed to be flexible and incrementally adoptable. Depending on your use case, Vue can be used in different ways by

2.2.3.1 Enhancing static HTML without a build step

2.2.3.2 Embedding as Web Components on any page

2.2.3.3 Single-Page Application (SPA)

2.2.3.4 Fullstack / Server-Side Rendering (SSR)

2.2.3.5 Jamstack / Static Site Generation (SSG)

2.2.3.6 Targeting desktop, mobile, WebGL, and even the terminal



Figure 5 [Vue.js](#)

2.2.4 [Node.js](#)

Node.js is a runtime environment that allows you to run JavaScript on the server side (not just in the browser). It is built on Chrome's V8 engine and is widely used for backend development.



Figure 6 [Node.js](#)

2.2.5 PostgreSQL and pgAdmin

2.2.5.1 PostgreSQL

PostgreSQL is a powerful, open source object-relational database system that uses and extends the SQL language combined with many features that safely store and scale the most complicated data workloads and comes with many features aimed to help developers build applications, administrators to protect data integrity and build fault-tolerant environments, and help you manage your data no matter how big or small the dataset.



Figure 7.1 PostgreSQL

2.2.5.2 pgAdmin

pgAdmin is a management tool for PostgreSQL and derivative relational databases such as EnterpriseDB's EDB Advanced Server. It may be run either as a web or desktop application. For more information on the features offered, please see the Features and Screenshots pages.



Figure 7.2 pgAdmin

2.2.6 Ultralytics YOLO26

YOLO26 is the latest release of the YOLO series. Unlike previous iterations, YOLO26 takes a deployment-first approach by streamlining the detection pipeline to deliver exceptional performance on edge devices, mobile platforms, and resource-constrained environments. This edge-optimized philosophy manifests through several innovations: the removal of Distribution Focal Loss (DFL), native end-to-end NMS-free inference, and the introduction of Progressive Loss Balancing (ProgLoss) combined with Small-Target-Aware Label Assignment (STAL). Powered by the novel MuSGD optimizer, YOLO26 achieves faster convergence during training while maintaining the accuracy and speed that have made the YOLO family the gold standard for real-time detection.

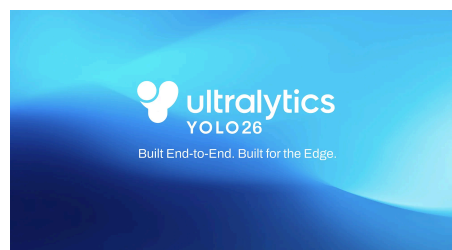


Figure 8 Ultralytics YOLO26

2.2.7 Python

Python is a high-level programming language that is very popular because it is easy to use and its code is quick to understand, unlike some more complex languages such as C or Java. It is suitable for both beginners and professionals and Can works on cross-platform with Windows, Linux, and macOS.



Figure 9 Python

2.3 Related Work

2.3.1 Define parking spaces using Ultralytics YOLO26

2.3.1.1 Capture a frame from a video or camera stream.

2.3.1.2 Use the provided code to launch a GUI for selecting an image and drawing polygons to define parking spaces.

2.3.1.3 Save the labeled data in JSON format for further processing.
For comprehensive instructions, check the selection of points section above.

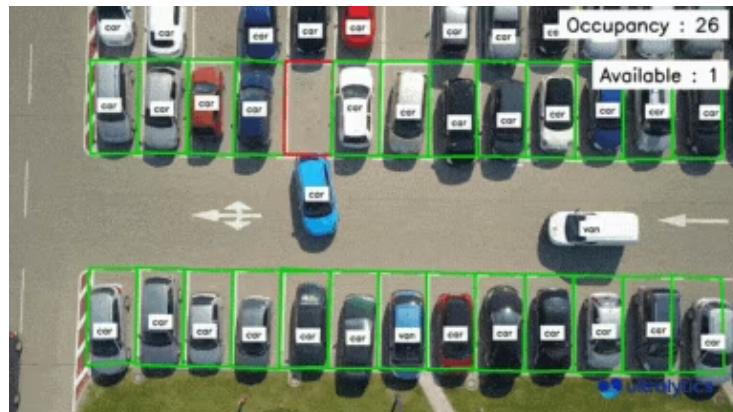


Figure 10 Define parking spaces using Ultralytics YOLO26

2.3.2 How LPR (License Plate Recognition) Cameras Work

2.3.2.1 License Plate Capture

When a vehicle enters the designated detection range, the LPR camera automatically captures an image of the license plate with high speed and precision. Regardless of how fast the vehicle is moving or the lighting conditions (day, night, glare), the camera is capable of producing a clear and readable image.

2.3.2.2 AI-Based Analysis

The captured plate image is then processed by intelligent software using OCR (Optical Character Recognition) and AI (Artificial Intelligence) technologies. The system analyzes and extracts characters, numbers, and even regional identifiers (such as province names) from the license plate with high accuracy.

2.3.2.3 Conversion to Digital Data

After analysis, the system converts the visual information into a digital format, such as: “ ๗๓ 1234 กรุงเทพมหานคร ” (example equivalent of a Thai plate). This structured data can then be easily used across multiple applications and systems.

2.3.2.4 Storage and Processing

The recognized license plate data is stored in a database along with timestamps and captured images. This allows for

- Historical tracking and verification
- Integration with access control systems (entry/exit management)
- Real-time alert systems (e.g., blacklist/whitelist detection)

2.3.3 The Recognition Pipeline

2.3.3.1 Face Detection

The system must first locate the face within a complex scene. This relies on object detection algorithms to separate the face from the background. State-of-the-art models like YOLO26 are frequently used in this stage to generate accurate bounding boxes in real-time.

2.3.3.2 Feature Analysis

Once the face is isolated, the software maps key nodal points, such as the distance between the eyes, the width of the nose, and the contour of the jawline. This process involves feature extraction to identify landmarks that remain consistent despite changes in lighting or expression.

2.3.3.3 Encoding

The analyzed geometry is converted into a numerical vector or "faceprint," often referred to as an embedding. This mathematical representation allows the computer to process facial data efficiently.

2.3.3.4 Matching

The system compares the new faceprint against a vector database of known individuals. If the similarity score exceeds a predefined confidence threshold, the identity is verified.

CHAPTER 3

ANALYSIS AND DESIGN

3.1 System design

The workflow of the Parking Management System is designed to streamline the user experience through automation and real-time data. The process is summarized as follows

3.1.1 User type

All users can access the system immediately. But if you have an MFU account, you can log in to the system. Login with the account will receive parking history. By default will be a guest user.

3.1.2 Real time Checking

Both user types proceed to display a building real-time map of parking slots. The system shows slot availability if a slot is occupied, the user is redirected to another parking slot. The parking slot will be checked by the parking line with YOLO26 and send data to the building map.

3.1.3 Parking time Collection

Once the vehicle enters a parking building, CCTV in the entrance lane will capture the license number and face recognition. When parking, the CCTV system will snapshot every 4 seconds. Parking slots will be detected for 2 scenarios,

1. Vehicles not parking will see a parking line.
2. Vehicles park will not see parking lines.

Parking vehicles will detect license numbers again to confirm parking. This module uses YOLO26 to recognize license plates numbers and face recognition. After processing this data will be stored, updated slots and kept in the database.

3.1.4 Data collecting

Captures all metadata (License numbers, face owner identity, parking slot number and entry time) is transmitted to the webpage and stored in the database.

3.1.5 Exit & History Management

When a vehicle exits the parking slot, the CCTV snapshot to find the parking line and license plate at vehicle exiting. After exit the parking slot, CCTV in the exit lane will detect the face for verification that the user has the same face as they entered.

The data that was detected will update the database. Guest users only complete the process and history log only staff can access in their, while MFU account users are received Parking history information details to view their parking logs.

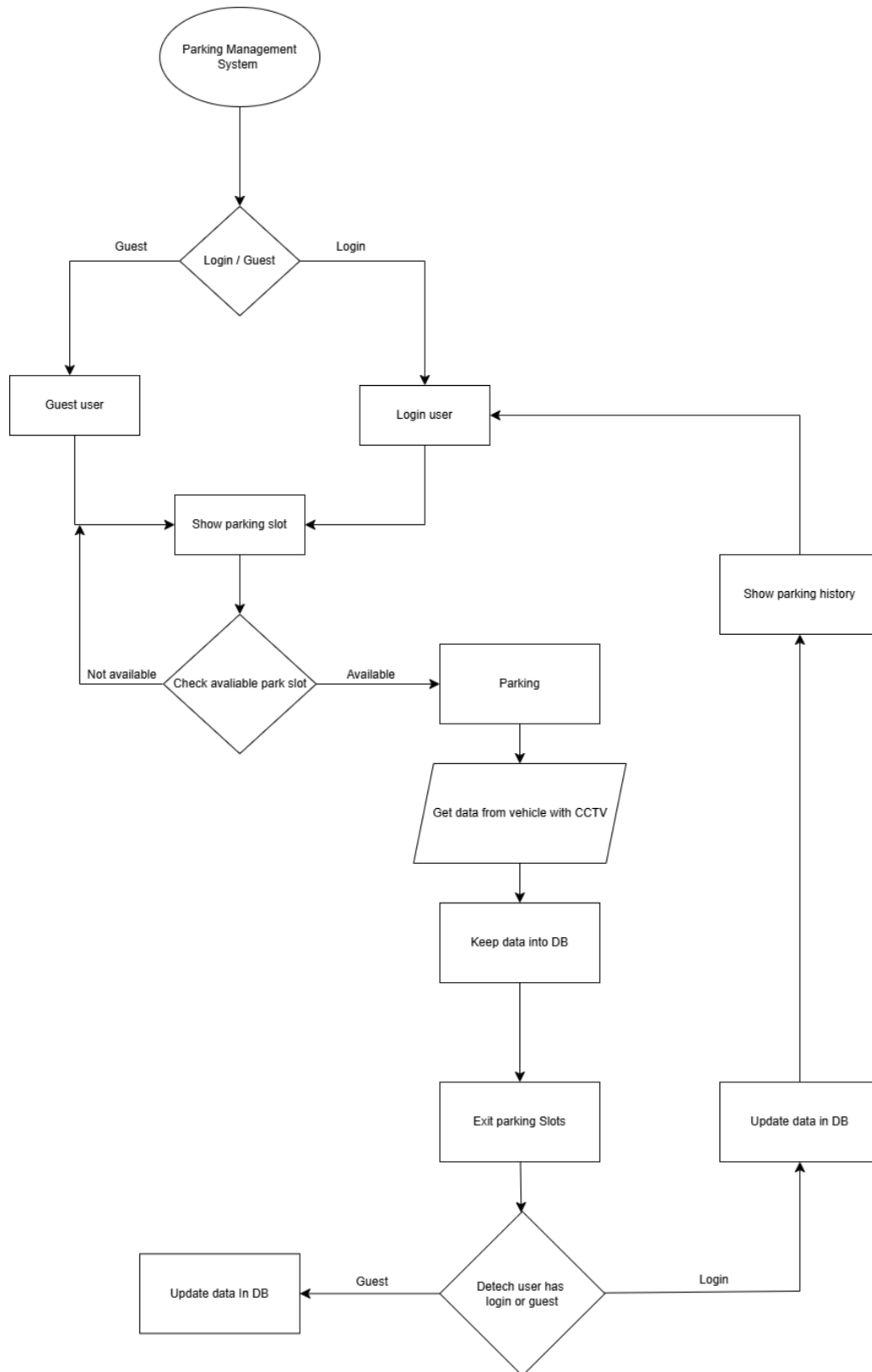


Figure 11 System Diagram

3.2 Use case diagram

3.2.1 User

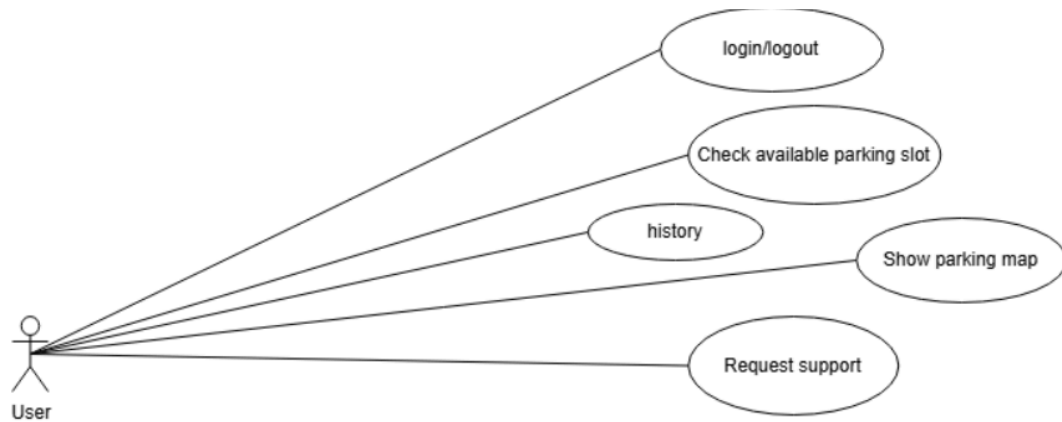


Figure 12 Use case diagram for user

3.2.2 Staff

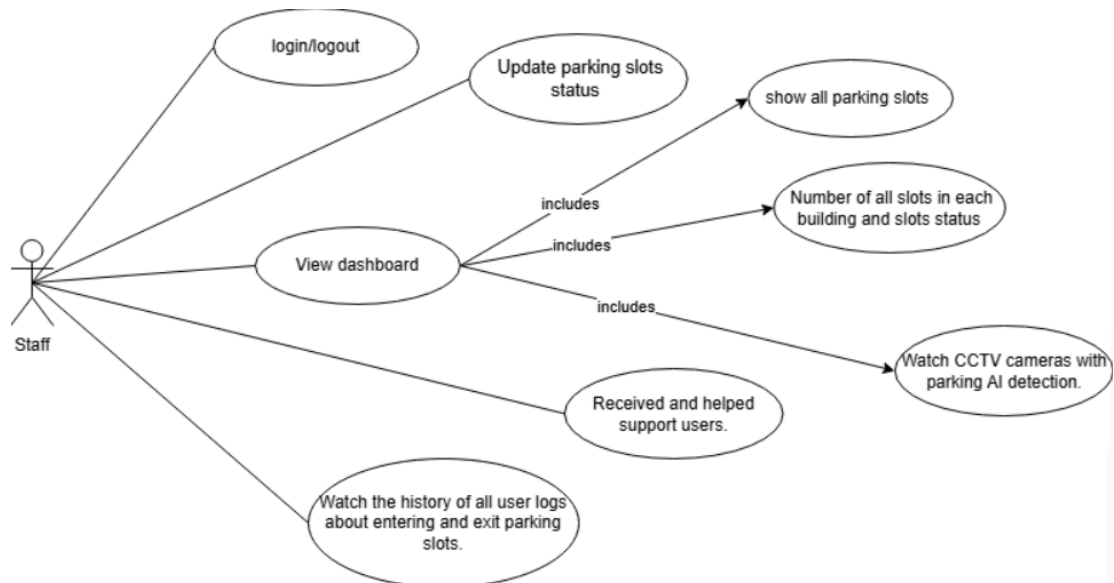


Figure 13 Use case diagram for staff

3.2.3 Admin

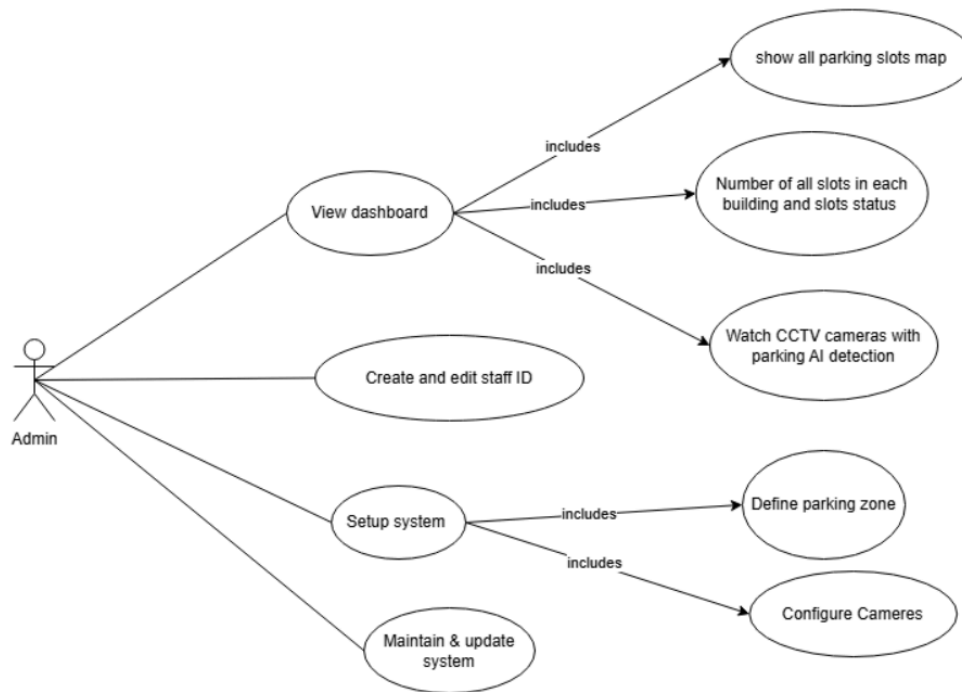


Figure 14 Use case diagram for admin

3.3 UI Design

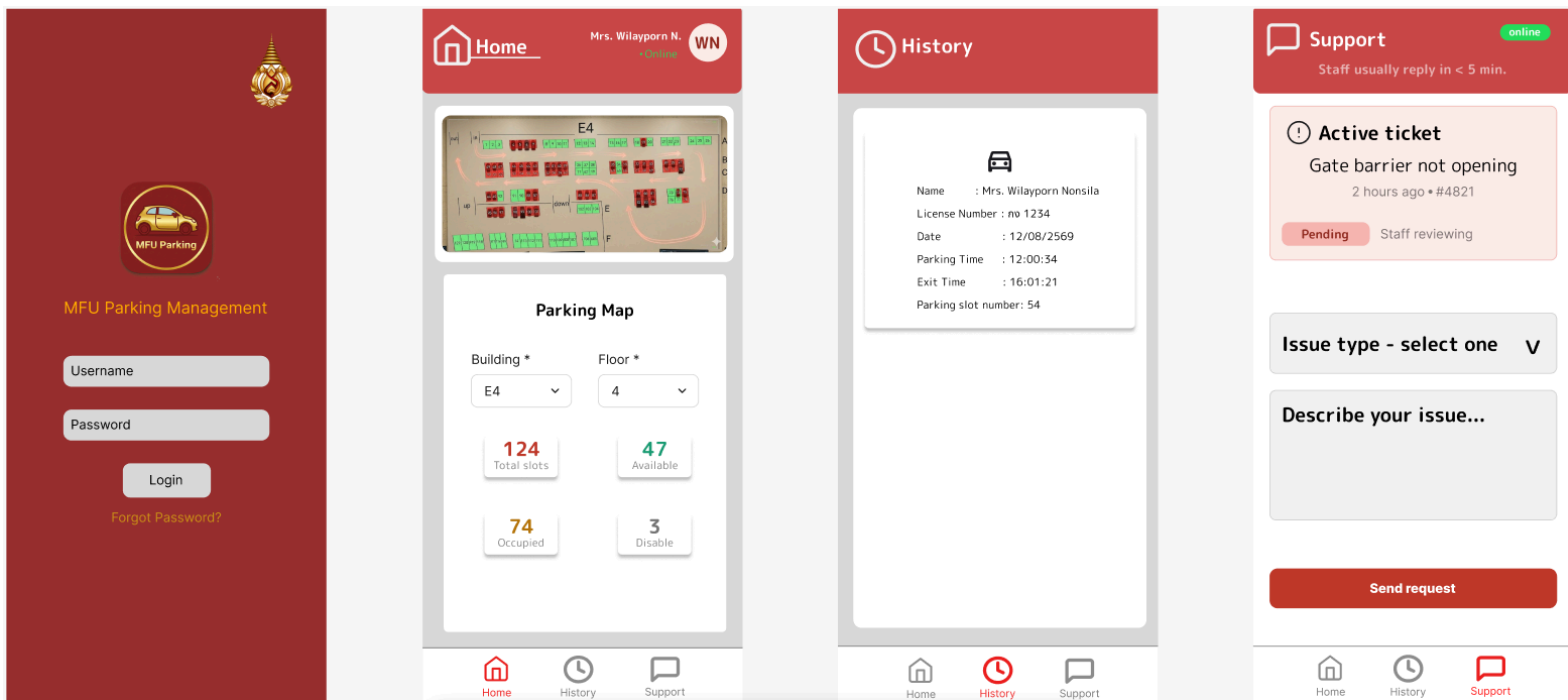


Figure 15 Parking map, History and Chat support for user

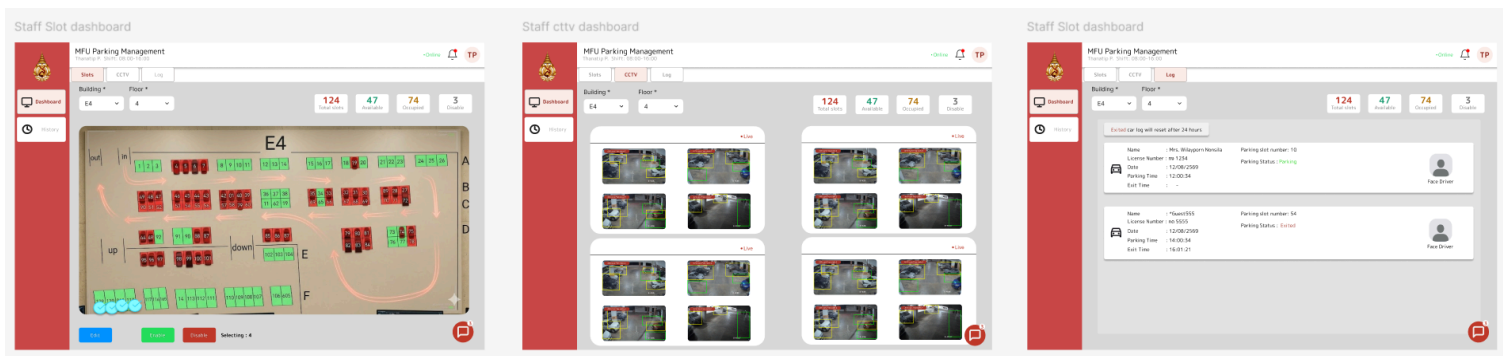


Figure 16 UI Design for dashboard in Staff

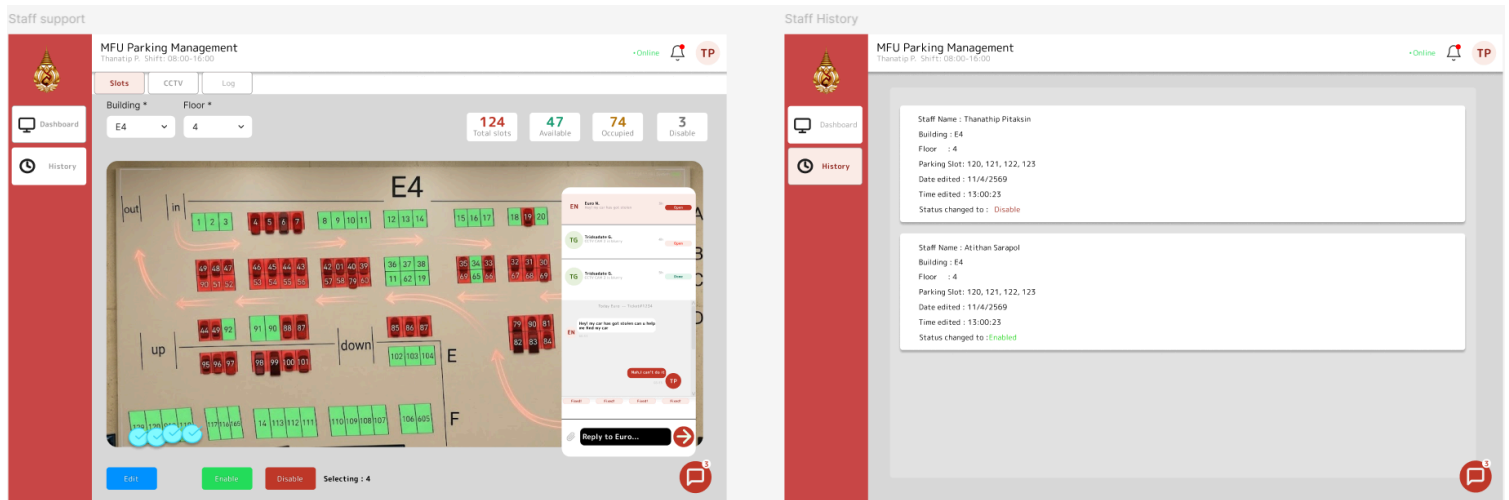


Figure 17 UI Design for Support and History in Staff

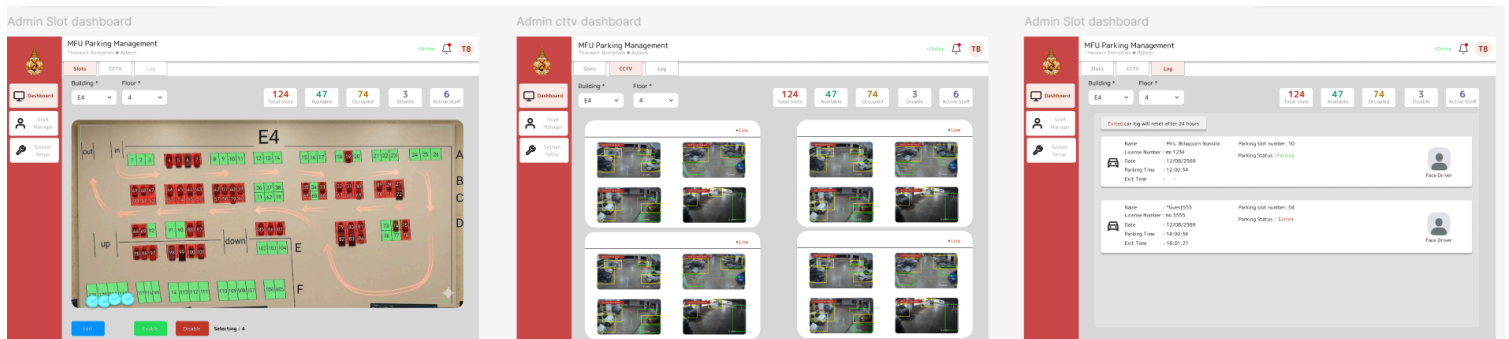


Figure 18 UI Design for History and Support in Admin

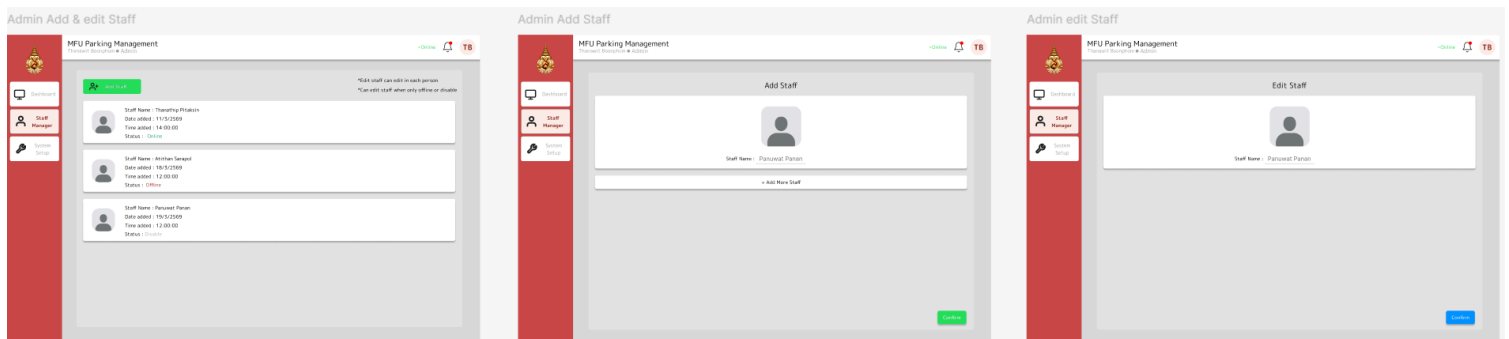


Figure 19 UI Design for Add & Edit in Admin

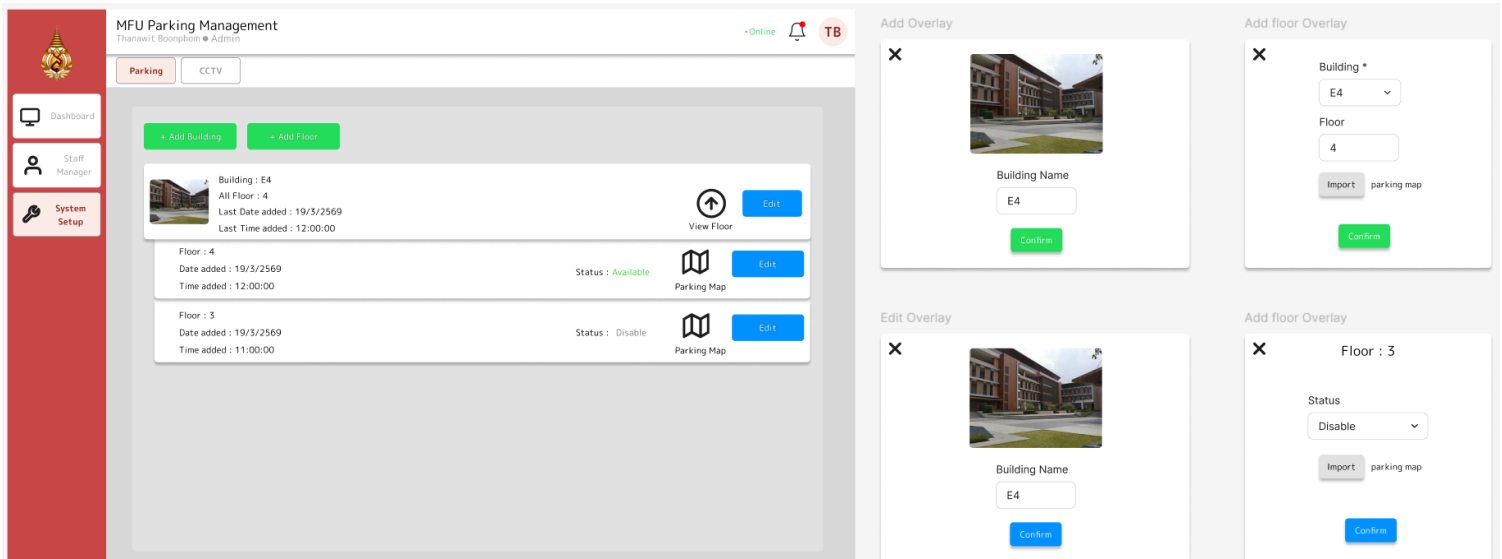


Figure 20 UI Design for Add or Edit Buildings & Floors in Admin

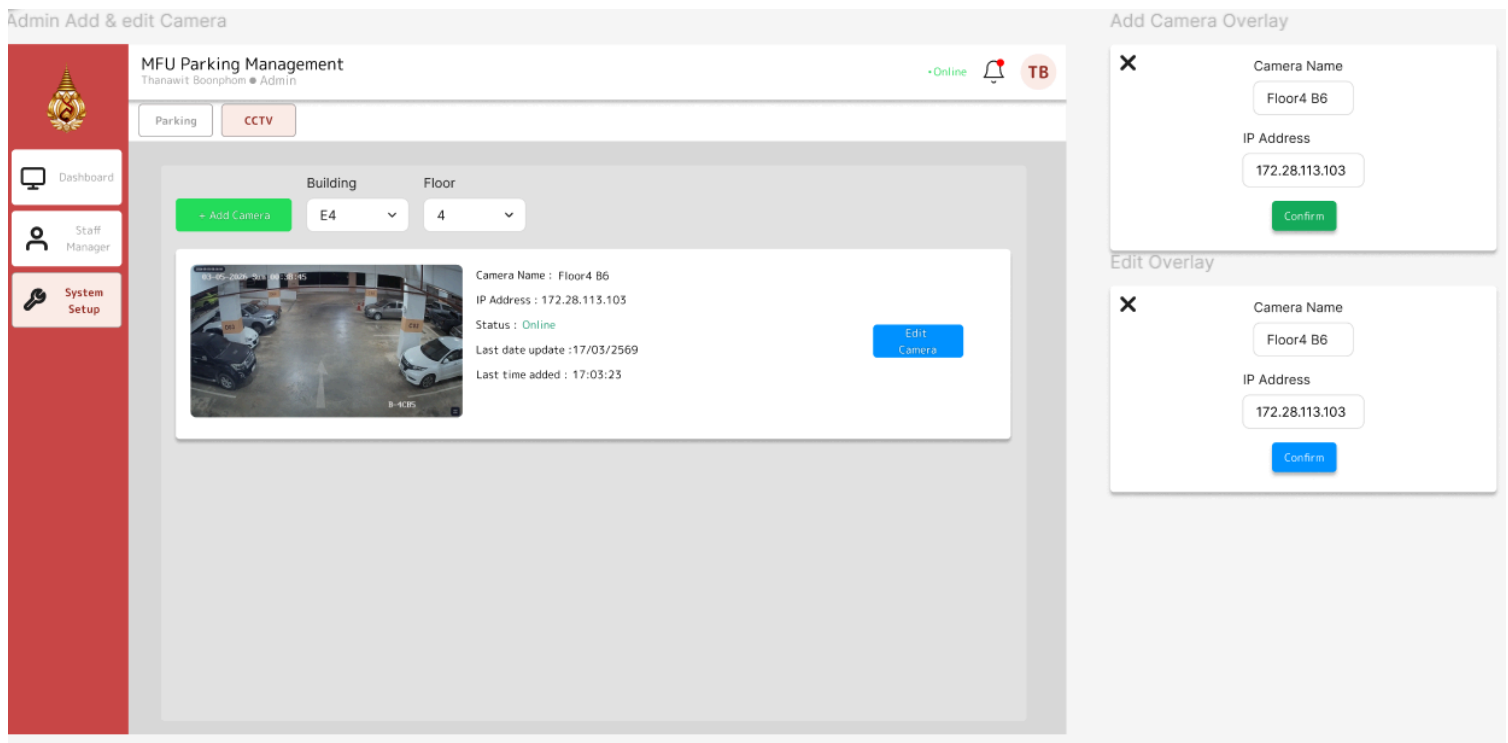


Figure 21 UI Design for Add & Edit Cameras in Admin

CHAPTER 4

CONCLUSION

Conclusion

Summarize what you have done in the senior project.

REFERENCES

- [1] “Visual Code Studio” [Online]. Available: <https://coderspace.io/en/blog/what-is-visual-studio-code-what-is-it-used-for/>
- [2] “Node.js” [Online]. Available: <https://nodejs.org/en/about/>
- [3] “PostgreSQL” [Online]. Available: <https://www.postgresql.org/about/>
- [4] “pgAdmin” [Online]. Available: <https://www.pgadmin.org/>
- [5] “Vue.js” [Online]. Available: <http s://vuejs.org/guide/introduction.html>
- [6] “Figma” [Online]. Available <https://www.figma.com/design/>
- [7] “YOLO26” [Online]. Available <https://datature.io/blog/yolo26-the-edge-first-evolution-of-real-time-object-detectio n>
- [8] “Parking Management using Ultralytics YOLO26” [Online]. Available https://docs.ultralytics.com/guides/parking-management/?fbclid=IwZXh0bgNhZW0CMTAAYnJpZBEuU0o5TjZMVzRFQjQyUVVJ3NHNdGMGYXBwX2lkEDlzMjAzOTE3ODgyMDA4OTIAAR5B3gnJly92q6FhNE_w56wdyyvq5u99GcDHApK51p0OfDtARx0f9i5QVV6_3g_aem_5S6o5nAqvzMLustAEfF1Bg
- [9] “Python” [Online]. Available <https://www.python.org/about/>
- [10] “LPR” [Online]. Available <https://www.gtoengineer.com/lpr-camera-guide/>
- [11] “Facial Recognition” [Online]. Available <https://www.ultralytics.com/glossary/facial-recognition>